

A Boundary-Complete Dyadic Prefix–Tail Path Archive

Exact Column Conservation and Fixed-Shift Provenance

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Abstract

TPC-133 supplies an executable native entrance but stops before the first block decomposition. We construct a complete growing path archive through the dyadic and divisor-prefix operations used in TPC-16 and TPC-17. An explicit normalized C^∞ bump gives $\sum_j \psi(x/2^j) = 1$ exactly. For every dyadic block b , a predeclared integer $D_0(b) \in [0, V]$ sends each opened divisor record to exactly one of $d \leq D_0(b)$ and $D_0(b) < d \leq V$. Thus every native column has outgoing weight one, the complete matrix satisfies

$$\mathbf{1}^T M_{134} = \mathbf{1}^T,$$

and native provenance, the prescribed shift, and the single physical normalization intertwine on every path. The resulting identity

$$B_{h_0, \delta}(X) = \sum_b (\mathcal{C}_b(D_0(b)) + \mathcal{T}_b(D_0(b), V))$$

is exact for every X . No prefix or tail estimate is inferred. The partition theorem is L0, its literal attachment is L1, and no positive L2 arithmetic saving is proved.

1 Input archive and an explicit smooth partition

Import the support-envelope native records

$$\alpha = (\ell, k, d), \quad c_X(\alpha) = -\Lambda(\ell)\mu(d)r_Q(\ell k + h_0)W(\ell k/X)$$

from TPC-133 [2]. They retain d rather than only the compressed pair (ℓ, k) .

Define

$$\chi(x) = \begin{cases} \exp\left(-\frac{1}{x-1/2} - \frac{1}{2-x}\right), & 1/2 < x < 2, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$Z(x) = \sum_{j \in \mathbb{Z}} \chi(x/2^j), \quad \psi(x) = \frac{\chi(x)}{Z(x)}.$$

For every $x > 0$, the sum defining $Z(x)$ is finite and positive. Moreover $Z(x/2^j) = Z(x)$, by reindexing.

Lemma 1.1 (Exact normalized dyadic partition). *The function ψ is smooth, compactly supported in $[1/2, 2]$, and*

$$\boxed{\sum_{j \in \mathbb{Z}} \psi(x/2^j) = 1} \quad (x > 0).$$

Only finitely many summands are nonzero for each x .

Proof. Smoothness at the endpoints is the standard flat-bump property. The support gives local finiteness. Since all denominators equal $Z(x)$,

$$\sum_j \psi(x/2^j) = \frac{\sum_j \chi(x/2^j)}{Z(x)} = 1.$$

□

This explicit formula removes an ambiguity in a machine archive: the program stores a symbolic ψ -node and the finite dyadic orbit. It does not approximate the column sum in binary floating point.

2 The prefix–tail compiler

Let $\mathcal{B}_X$ be the finite set of dyadic pairs

$$b = (j_L, j_K), \quad L = 2^{j_L}, \quad K = 2^{j_K},$$

which meet at least one native atom. Fix before inspecting any coefficient a deterministic policy

$$D_0 : \mathcal{B}_X \longrightarrow \{0, 1, \dots, V\}.$$

The exact theorem below works for every such policy. The bundled certificate uses the maximal integer satisfying a separately declared published-core screen, with value zero when the screen is empty.

Definition 2.1 (Path children). For $\alpha = (\ell, k, d)$ and $b = (j_L, j_K)$, create one child when

$$\psi(\ell/L)\psi(k/K) \neq 0.$$

Its edge multiplier is

$$m(\alpha, b) = \psi(\ell/L)\psi(k/K),$$

and its terminal type is

$$\text{type}(\alpha, b) = \begin{cases} \text{PREFIX}, & d \leq D_0(b), \\ \text{TAIL}, & D_0(b) < d \leq V. \end{cases}$$

Theorem 2.2 (Boundary-complete path conservation). *For every native atom,*

$$\sum_{b \in \mathcal{B}_X} m(\alpha, b) = 1.$$

The prefix and tail types are disjoint and exhaustive, including the boundary $d = D_0(b)$. Consequently the complete path matrix M_{134} satisfies

$$\boxed{\mathbf{1}^T M_{134} = \mathbf{1}^T.}$$

Proof. By [lemma 1.1](#),

$$\sum_{j_L, j_K} \psi(\ell/2^{j_L})\psi(k/2^{j_K}) = \left(\sum_{j_L} \psi(\ell/2^{j_L}) \right) \left(\sum_{j_K} \psi(k/2^{j_K}) \right) = 1.$$

Compact support makes the sum finite. For each block and each $d \leq V$, exactly one of $d \leq D_0(b)$ and $d > D_0(b)$ holds. Thus no second multiplier is introduced by the type split. □

Theorem 2.3 (Metadata intertwining). *Suppose every child copies its native tuple, h_0 , and the inherited physical normalization. Then each of these coordinate projections intertwines with M_{134} . In particular no averaged shift or second normalization is created by the block compiler.*

Proof. Every nonzero edge has equal source and target metadata. The entrywise criterion of TPC-123 applies [\[1\]](#). □

3 Exact scalar block identity

For a block b , define

$$\begin{aligned}\mathcal{C}_b(D_0(b)) &= \sum_{\ell, k} \sum_{\substack{d|k \\ d \leq D_0(b)}} c_X(\ell, k, d) \psi(\ell/L) \psi(k/K), \\ \mathcal{T}_b(D_0(b), V) &= \sum_{\ell, k} \sum_{\substack{d|k \\ D_0(b) < d \leq V}} c_X(\ell, k, d) \psi(\ell/L) \psi(k/K).\end{aligned}$$

Corollary 3.1 (Growing prefix–tail archive identity). *For every X ,*

$$B_{h_0, \delta}(X) = \sum_{b \in \mathcal{B}_X} (\mathcal{C}_b(D_0(b)) + \mathcal{T}_b(D_0(b), V)).$$

Proof. Apply [theorem 2.2](#) to the native coefficient vector and then partition the opened divisors at $D_0(b)$. This is the exact coefficient split used in TPC-17 [\[3, 4\]](#). $\square$

Remark 3.2 (No parameter-transport shortcut). TPC-16 also proves that two complete Vaughan hard packets can differ by $O_A(X(\log X)^{-A})$. It explicitly warns that this total signed packet equivalence does not match individual factor pairs. We do not insert it as a native-column bijection. A proof-carrying transport would have to archive its soft row terms separately.

4 Executable certificate and limits

The standard-library program `experiments/tpc134_branch_archive.py` reads the committed TPC-133 JSONL records, computes every finite dyadic orbit, applies one canonical D_0 policy, and writes a path JSONL plus a summary certificate. It rejects a deleted or duplicated overlap child, a prefix/tail boundary mutation, a wrong source, a wrong shift, and a second normalization. The certificate checks the exact finite index set and records column conservation as an invocation of [lemma 1.1](#); it does not replace that symbolic identity by a floating-point column-sum test. Its `-check` mode is read-only.

Statement	Status
Smooth dyadic identity and divisor split	Exact L0.
Complete native-to-block path archive	Proved at this cut, L1.
Uniform smallness of prefix blocks	Not invoked in this paper.
Uniform smallness of tail blocks	Not proved.
Physical/determinant/zero-mode total maps	Not supplied.
Positive fixed- h_0 arithmetic power saving	Not proved; no positive L2.

Kill criteria. A nonzero column defect, a gap or overlap at $d = D_0$, a lost source, or a scope mutation is STOP-DECLARED-ROUTE for the declared compiler. A missing later carrier map is NOT-TESTABLE, not a contradiction.

Explicit exclusions. This paper proves no determinant-energy lower bound, zero-mode estimate, physical cover, signed affine cancellation, endpoint below $1/400$, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem.

References

- [1] Liang Wang. Literal native-atom path archives for the tpc-15 hard packet. *Preprint*, 2026.
- [2] Liang Wang. An executable native entrance for the tpc-15 hard packet. *Preprint*, 2026.
- [3] Liang Wang. Square-root normal forms for a fixed-shift type-ii gate. *Preprint*, 2026.
- [4] Liang Wang. Assembly and concentration at a fixed-shift type-ii gate. *Preprint*, 2026.